Your Camera Roll, Your Story: Selection and Storytelling with Adjustable AI Help

Kashyap Shankar Iyer (3087617) · Intelligent Interactive Systems, Assignment 1: Literature Review

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Problem and context

After a trip, people hold hundreds of photos they want to share. A social media post asks for two things: choosing a few photos and writing words that fit them.

Generative AI can now help with both, and how much it writes can vary. Draxler et al. compared four ways of producing a postcard from a trip: writing it themselves, editing AI text, choosing between AI texts, or getting a text the AI wrote in full [2]. A personal post is how people present themselves, so it matters where on this range a post sits. This review asks how AI can help a person turn a trip album into posts when the person chooses how much of the writing the AI does. It is relevant to anyone sharing personal memories through AI tools.

Related work

Telling stories with photos (Figure 1). Balabanović et al. found that people tell photo stories in two ways: *photo-driven*, where each photo prompts the story, and *story-driven*, where the story leads and photos are found to fit it [1]. Their participants told stories aloud with a prototype device, so the study shows that story-driven telling happens, not how AI could support it.

AI writes from photos. VIST, a visual storytelling dataset, pairs 20,211 photo sequences with crowd-written stories so models can learn to generate them [5]. Because crowd workers wrote these stories about other people's photos, the target is a typical story, not the voice of the person who took them.

Co-writing and ownership. In Draxler et al.'s study, ownership was low for AI-generated text and rose with the person's influence over it, yet people rarely declared AI authorship [2]. An opinionated assistant shifted both what participants wrote in a social media post and their later views [6]. Vistoria, which combines text and image editing for fiction, measurably raised writers' workload, and most of its 12 participants said in interviews that they felt more ownership than with a text-only baseline [3]. Together these studies show that AI help changes both how far people feel a text is theirs and what it says.

Design principles. Mixed-initiative systems should account for uncertainty and make correction cheap [4]. In Kulesza et al.'s study, explanations users could act on helped them correct a classifier up to twice as efficiently [7]. Together they suggest that each match should come with a checkable reason and a way to swap it.

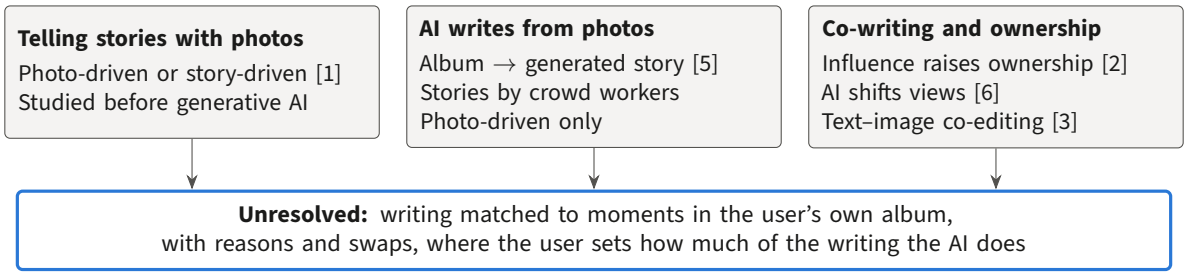


Figure 1: Literature map: three strands of related work and the question none of them addresses.

Gap or opportunity

The AI work on photos starts from the photos and generates words for them. Yet Balabanović et al. found that people sometimes start from the story they want to tell and then look for photos that fit it [1]. In the work I reviewed, no system lets a person set how much of a post's writing the AI does, from AI-written to entirely their own, and then matches that writing to moments in their own

album, meaning groups of photos taken close together. The co-writing studies vary AI involvement in text-only or fictional tasks; none covers the end of the range where the AI chooses the photos and the person writes every word. It is unknown whether people find such matches meaningful, how ownership changes across the range when the photos are their own, whether a short reason per match makes correction easier, as found for other tasks [7], and whether people correct matches at all, given that AI suggestions changed what participants wrote [6]. It is also unclear how well a model can match a line such as "I didn't want the day to end", which describes a feeling rather than anything visible in a photo.

Possible contribution

I propose this question: when a system matches writing to moments in a person's own trip album, with a one-line reason and a swap for each match, how do match quality and ownership change as the writing moves from AI-written, to AI suggestions the person accepts or rejects, to entirely their own? Outcomes would be perceived match quality, sense of ownership (adapting the measures of Draxler et al. [2]) and willingness to post.

The intended application is a tool for turning a trip album into social media posts, where the person decides how much of the writing the AI does and can see which words are the AI's, since people rarely declare AI authorship themselves [2].

Initial limits: the scope is trip albums and short writing; a first study would be small and use participants' own photos, raising consent and privacy issues; and weak matching on abstract lines may limit what it can show.

References

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